

Criteria	Submission Requirements
Produce classes that represent near-Earth objects and their close approaches. (Task 1)	The NearEarthObject class represents a near-Earth object. - The constructor assigns attributes for: - designation: The NEO's primary designation. - name: The NEO's IAU name (could be empty, or None) - diameter: The NEO's diameter, in kilometers, or NaN. - hazardous: Whether the NEO is potentially hazardous - approaches: A collection of this NEO's CloseApproaches (initially an empty collection). The CloseApproach class represents a close approach to Earth by an NEO. - The constructor assigns attributes for: - time: The date and time, in UTC, at which the NEO passes closest to Earth.

	• distance: The nominal approach distance, in astronomical units, of the NEO to Earth at the closest point. • velocity: The velocity, in kilometers per second, of the NEO relative to Earth at the closest point. • neo: A reference to the NearEarthObject that is making the close approach (initially None). • An additional attribute, to store the NEO's primary designation before the CloseApproach is linked to its NearEarthObject Additionally, each of these classes should implement a <code>__str__</code> method that produces a human-readable description of the contents of the object.
Load data from CSV and JSON files into Python objects. (Task 2a)	The <code>load_neos</code> function loads NEO data from a CSV file. • The function opens the given file for reading. • The function uses the <code>csv</code> module to parse the file

contents into a standard Python data structure (e.g. list, dict, etc).

- The function converts this raw data into a collection of NearEarthObjects

- The function returns a collection of NearEarthObjects

The load_approaches method loads close approach data from a JSON file.

- The function opens the given file for reading.

- The function uses the json module to parse the file contents into a dict.

- The function converts this raw data into a collection of CloseApproach objects.

- The function returns a collection of CloseApproach objects.

Data from the extraneous columns (CSV) and fields (JSON) shouldn't be bound to the constructed NearEarthObjects and CloseApproaches.

Link together Python objects and any required metadata in a

The NEODatabase constructor captures and preprocesses a

database wrapper that supports basic inspection. (Task 2b)

collection of NEOs and close approaches.

- The constructor captures and saves its arguments, a collection of NearEarthObject and a collection of CloseApproaches.
- The constructor precomputes auxiliary data structures to assist with the get_neo_by_designation and get_neo_by_name methods.
- At the end of the constructor, the .neo attribute is set on each close approach to the matching NearEarthObject.
- At the end of the constructor, the .approaches attribute is populated for each NearEarthObject with a collection of its close approaches.

Style (Mechanics)

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Produce Python code that satisfies PEP 8.	Submitted code follows the guidelines of PEP 8 - the Style Guide for Python.

Write Python comments that satisfy PEP 257.	Submitted code follows the docstring conventions of PEP 257. Each module contains a module-level docstring describing the purpose of the module. Complex functions, classes, and methods include a docstring that: • Starts with the primary action of the callable in the imperative mood • Includes any additional clarifications needed • Provides descriptions of parameters and return values
Submit code free from starter code markings.	There are no <code># TODO</code> comments left in the submitted code. Portions of comments that say <code># ELABORATE</code> have been filled in with a description of the corresponding code.

Style (Design)

Criteria	Submission Requirements
Divide attributes appropriately among the <code>NearEarthObject</code> class and the <code>CloseApproach</code> class.	Attributes of <code>NearEarthObjects</code> and <code>CloseApproaches</code> are captured in the constructor from the supplied arguments.

	Instances of NearEarthObject don't have attributes of individual close approaches. Instances of CloseApproach don't have attributes of the associated NEO (except for the primary designation needed to initially link the close approach to its NEO).
Attach functionality to classes only when appropriate.	Standalone functions are used when the functional operation doesn't depend on external state and does not conceptually belong on an object.
Design a coherent system to filter objects by user-specified criteria.	The logic backing the filter system is consistent and doesn't contain excess duplicated code.
Consume and produce streams of data as iterables.	Represents concrete data (buffered file contents, static collections of NEOs or close approaches, auxiliary data structures) as concrete. Represents streaming data (close approaches that match criteria, limited stream of results) as streaming.

